
Fashion Compatibility Learning via Metric Learning and Modern Vision Backbones

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Abstract

A major challenge when building an outfit is knowing how well clothing items go together. We address this problem as a metric learning task: given images of individual clothing items (tops, bottoms, shoes), we learn visual embeddings such that compatible items—those belonging to the same curated outfit—are mapped nearby in the learned space. Using the Polyvore dataset, we compare two modern vision backbones (Swin Transformer and ConvNeXt) fine-tuned from ImageNet pretraining under triplet and contrastive losses. Retrieval is performed via k -Nearest Neighbors on the learned features. Our central goal is to investigate whether these modern, general-purpose architectures can learn sufficient visual compatibility signals without relying on explicit type conditioning or category-aware supervision, as required by prior state-of-the-art methods.

1 Introduction and Background

Choosing items that go well together is a non-trivial perceptual task: visual compatibility depends on subtle cues such as color harmony, texture pairing, and stylistic coherence. Automating this judgment has direct applications in e-commerce recommendation and virtual styling tools.

Problem statement. We frame compatibility as a metric learning problem. Given a collection of clothing item images $\{x_1, \dots, x_n\}$ spanning categories (tops, bottoms, shoes), we learn an embedding function $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ such that items co-appearing in the same outfit are mapped nearby while items from different outfits are pushed apart:

$$d(f_\theta(x_i), f_\theta(x_j)) \ll d(f_\theta(x_i), f_\theta(x_k)) \quad (1)$$

where (x_i, x_j) is a compatible pair and x_k is an incompatible item. At inference, compatibility-aware retrieval is performed with k -NN over the embedding space.

Related work. Vasileva et al. [2018] proposed type-aware embeddings (ECCV 2018) which jointly learn item similarity and compatibility. Their key insight is that compatibility between two items depends on their types (e.g., top-bottom vs. top-shoes), and they learn type-specific projection subspaces from a shared general embedding. Trained on their Polyvore Outfits dataset (68,306 outfits), this method yields 3–5% improvement over prior work on fill-in-the-blank and compatibility prediction. However, it requires explicit type labels at both train and test time, limiting generalization to novel categories.

Tan et al. [2019] (ICCV 2019) address this limitation by proposing a model that learns the different similarity conditions—such as color, category, or shape—as latent variables, without explicit supervision. Their method outperforms condition-supervised approaches on Polyvore-Outfits, Maryland-Polyvore, and UT-Zappos50k, and generalizes to unseen categories. This work is the most direct inspiration for our goal: we similarly aim to avoid conditioning on explicit type labels.

Sarkar et al. [2023] (WACV 2023) take a different approach with OutfitTransformer, which models compatibility at the outfit level. Using task-specific tokens and self-attention over all items in an outfit, it encodes full-outfit compatibility for prediction and complementary item retrieval. While powerful, this architecture requires the full outfit context as input, making it less suited to pairwise retrieval from a single query item—the setting we focus on.

Our goal and novelty. None of the above works examines whether modern backbone architectures—Vision Transformers and new-generation CNNs—reduce the need for type conditioning or explicit supervision when trained with metric learning on outfit groupings. We directly test this hypothesis by comparing Swin Transformer Liu et al. [2021] and ConvNeXt Liu et al. [2022] under triplet and contrastive losses, without any type-aware subspaces.

Contributions.

- We benchmark Swin-T and ConvNeXt-T backbones under a unified metric learning pipeline on Polyvore.
- We compare triplet margin loss and contrastive loss under identical experimental conditions.
- We measure the gap to type-aware methods Vasileva et al. [2018] and the unsupervised condition baseline Tan et al. [2019], quantifying how much supervision modern architectures can replace.

2 Method

2.1 Model Architecture

We use a shared-weight encoder f_θ to project all item images into a common embedding space $\mathbb{R}^d$ ($d = 128$ or 256), regardless of item category. Two pretrained backbones are compared:

Swin Transformer (Swin-T) Liu et al. [2021]: A hierarchical Vision Transformer that processes image patches in shifted local windows. It offers the representational power of attention while being efficient through its windowed design.

ConvNeXt-T Liu et al. [2022]: A modernized convolutional architecture that adapts Transformer design choices (depthwise convolutions, inverted bottlenecks, LayerNorm) to a standard CNN. It retains spatial inductive bias while matching or exceeding ViT performance on visual tasks.

Both backbones are pretrained on ImageNet-1K and followed by a two-layer MLP projection head with ReLU, producing ℓ_2 -normalized embeddings.

2.2 Metric Learning Objectives

We compare two standard metric learning losses, constructed from outfit groupings.

Triplet Margin Loss. Given anchor x_a , positive x_p (same outfit), and negative x_n (different outfit, semi-hard sampled):

$$\mathcal{L}_{\text{triplet}} = \max(0, \|f_\theta(x_a) - f_\theta(x_p)\|_2 - \|f_\theta(x_a) - f_\theta(x_n)\|_2 + \alpha) \quad (2)$$

with margin $\alpha = 0.3$.

Contrastive Loss. Given a pair (x_i, x_j) with label $y = 1$ if compatible:

$$\mathcal{L}_{\text{contr}} = y d_{ij}^2 + (1 - y) \max(0, m - d_{ij})^2, \quad d_{ij} = \|f_\theta(x_i) - f_\theta(x_j)\|_2 \quad (3)$$

with margin $m = 1.0$.

2.3 Training Setup

- **Dataset:** Marqo/polyvore on Hugging Face Hub, a subset of the Polyvore benchmark introduced by Vasileva et al. [2018]. It contains fashion item images grouped into outfits with category labels, item descriptions, and outfit groupings. Compatible pairs are drawn from outfit groupings; negatives are randomly sampled from other outfits. Standard train/val/test splits.
- **Preprocessing:** Resize to 224×224 , normalize with ImageNet statistics. Augmentations at train time: random horizontal flip, color jitter.
- **Optimizer:** AdamW with differential learning rates: 10^{-4} for the projection head, 10^{-5} for the backbone. Weight decay 10^{-4} . Batch size 64, up to 30 epochs with early stopping on validation AUC.

2.4 Retrieval

At inference, embeddings are extracted for all test items. Given a query item, we retrieve the k most compatible candidates by cosine-similarity KNN. If dataset scale requires it, we will explore approximate nearest neighbor search (e.g., FAISS).

3 Results

Note: This is a project proposal. The table below describes our planned evaluation; results will be filled in upon experimental completion.

Evaluation metrics. Following Vasileva et al. [2018] and Tan et al. [2019], we report: (i) **AUC** for binary compatibility prediction (compatible vs. random pair); (ii) **FITB accuracy** (Fill-in-the-Blank: select the correct missing item from 4 candidates); (iii) **Recall@K** for retrieval.

Table 1: Planned comparison on the Polyvore test set. Baselines from the literature are shown for reference.

Method	AUC	FITB Acc.	Recall@10
ConvNeXt-T + Triplet (ours)	—	—	—
ConvNeXt-T + Contrastive (ours)	—	—	—
Swin-T + Triplet (ours)	—	—	—
Swin-T + Contrastive (ours)	—	—	—
Type-Aware Vasileva et al. [2018] (supervised)	~ 0.88	$\sim 57\%$	—
Unsup. Conditions Tan et al. [2019]	~ 0.91	$\sim 67\%$	—

We will analyze the gap between our type-agnostic approach and the baselines, and inspect embedding geometry via t-SNE to understand whether category information spontaneously clusters, as observed by Vasileva et al. [2018].

4 Conclusion

We propose a study of fashion compatibility learning using modern vision backbones and metric learning in a type-agnostic setting. Our approach is directly motivated by the line of work from Vasileva et al. [2018] and Tan et al. [2019]: we ask whether the architectural improvements in Swin Transformer and ConvNeXt reduce the need for explicit type conditioning, moving further toward the unsupervised direction explored by Tan et al. Our findings are intended to inform whether backbone choice is a viable lever for reducing supervision requirements in structured visual compatibility tasks.

- **Anticipated outcome:** Quantitative comparison of Swin-T vs. ConvNeXt and triplet vs. contrastive loss on Polyvore compatibility benchmarks.
- **Key limitation:** Without type conditioning, the encoder may conflate category proximity with style compatibility. Embedding analysis will help characterize this.

References

- Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Rohan Sarkar, Navaneeth Bodla, Mariya I. Vasileva, Yen-Liang Lin, Anurag Beniwal, Alan Lu, and Gérard Medioni. OutfitTransformer: Learning outfit representations for fashion recommendation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 3590–3598, 2023.
- Reuben Tan, Mariya I. Vasileva, Kate Saenko, and Bryan A. Plummer. Learning similarity conditions without explicit supervision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- Mariya I. Vasileva, Bryan A. Plummer, Krishna Dusad, Shreya Rajpal, Ranjitha Kumar, and David Forsyth. Learning type-aware embeddings for fashion compatibility. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 390–405, 2018.